Components of AI Project Framework

The AI Project Cycle: from Problem Scoping to Deployment

Your Name

Your School / Institute

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Contents

Learning Outcomes

Students will be able to:

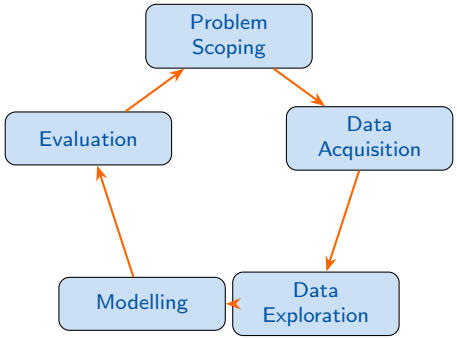
- know about the AI Project Cycle and appreciate its importance
- understand the nature of problem scoping
- set goals for an AI project
- explore the process of data acquisition
- comprehend data exploration using data visualisation techniques
- gain awareness about modelling and its approaches
- get familiar with evaluation and deployment



AI Project Cycle

What Is the AI Project Cycle?

- Like the water cycle, it shows the progression of a process.
- A **structured, step-by-step framework** for creating an AI system.
- Helps us define the problem, set goals, collect data, find patterns, and test the model.
- Each stage needs careful planning and execution.





The Stages of the Cycle

Problem scoping Understand and define the problem; set clear goals and objectives.

Data acquisition Collect relevant data from reliable, authentic sources in sufficient quantity.

Data exploration Analyse the data to find patterns, trends, and relationships, using visual representations.

Modelling Select an AI model that can learn from the data and make predictions.

Evaluation Test the model and compare results with expected outcomes.

Deployment

Once evaluation gives accurate results, the model can be deployed in the real world.

Activity 11.1: Museum Security Scenario

Scenario

You are the Chief Security Officer protecting high-value ancient artefacts that have been brought to a museum for exhibition. They are of great historical significance, and collectors worldwide want them.



- Information is circulating that a notorious art collector has recruited a master criminal to steal them.
- **Task:** enhance the museum's security and prevent any theft.
- **Follow-up:** explain the AI project cycle using a suitable example.



Problem Scoping and Goals

Problem Scoping

- Choosing a problem to address using AI.
- Identify the problem's scope, objectives, and constraints to get a clear understanding of what needs to be solved.
- Select a problem from a theme, an SDG-related issue, or one of your own.

Theme	Topics / Challenges
Smart Cities	Energy efficiency, traffic and waste management
Social Media	Sentiment analysis, fake news detection
Environmental Sustainability	Biodiversity, climate action, pollution
Health	Medicinal aid, spread of diseases
Agriculture	Pest issues, yield rates



More Themes (Activity 11.2)

Theme	Topics / Challenges
Vaccination Drive	Vaccine awareness, limited supply, vaccination centres, training healthcare workers
Hostel	Limited space, surveillance, cleanliness, rent collection, complaints, maintenance
School Attendance	Connectivity, hardware and software faults, user adoption, cost, inaccurate records

Activity 11.2

(1) Pick a theme and say why. (2) List problems related to it. (3) Draw a mind map of the problems.

Goal of a Project

- Defining the goal is one of the most important steps in project planning.
- A clear aim gives clarity and direction to all later actions.
- Defining a goal is iterative and needs cooperation, research, and careful evaluation.
- A well-defined goal guides decision-making and ensures the AI solution addresses real-life challenges.



4Ws Problem Canvas: Who and What

Problem scoping = identifying a problem, the factors affecting it, and the project goal. The canvas has four blocks: **Who? What? Where? Why?**

Who?

- Who are the stakeholders (directly or indirectly affected)?
- What do you know about them?

Stakeholders are involved in the problem and benefit from the solution.

What?

- What is the problem?
- How do you know it is a problem?

Gather evidence: observations, newspapers, social media posts, reports.

4Ws Problem Canvas: Where and Why

Where?

- In what context or location do stakeholders experience the problem?
- Physical places (cities, areas), online platforms, institutions, public spaces
- Local or global impact

Why?

- Why is the problem worth solving?
- How will the solution benefit stakeholders and society?
- Better quality of life, efficiency, cost savings, safety, sustainability

The canvas defines the problem, identifies stakeholders, specifies context, and highlights why solving it matters.

Problem Statement Template

- Summarises all key points of the 4Ws canvas in a single template.
- Lets us quickly recall the ideas later.
- Makes the important aspects easy to understand and remember for analysis and decision-making.



Example: Gopalapuram Air Pollution (Who & What)

Gopalapuram is a small urban area known for a clean, green environment, but rapid industrialisation, construction, and rising vehicular traffic have changed that.

Who?

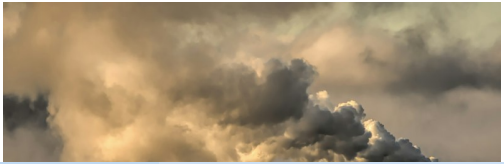
Residents, local government, environmentalists, healthcare authorities, animals and birds, industrialists.

- Residents: health and quality of life directly affected
- Government: environmental and public-health policy
- Environmentalists: protect biodiversity

What?

- Rising air pollution (industrial and vehicular emissions)
- More respiratory problems in citizens
- Insect and bird populations affected

Evidence: increased pollution; more respiratory illness.



Example: Gopalapuram Air Pollution (Where & Why)

Where?

Felt in daily life across residential areas, public places, workplaces, green spaces, educational institutions, and natural habitats.



Why? (value to stakeholders)

- Lower pollution and health risks for residents
- Lower healthcare burden
- Sustainable development for industry and authorities
- Preserved biodiversity and green spaces

How the situation improves

- Cleaner, more enjoyable living environment
- A sustainable, environmentally conscious city
- Ecological balance for insects and birds

Data Acquisition

What Is Data? Training vs. Testing Data

Data is a collection of facts or statistics gathered for reference or analysis (text, video, images, audio, ...). Data is the fuel for AI algorithms.

Training data

Initial dataset used to train the model. It must be aligned with the problem statement and be sufficient, relevant, accurate, and wide-ranging.

Testing data

Data the algorithm has *not seen before*, used to evaluate performance. It should represent what the model will meet in the real world.



Data Features and the Cricket Example

- **Data features** describe the type of information to collect in response to the problem statement.
- Visualise the factors that influence the problem and find parameters that affect it directly or indirectly.

Predicting the result of a match between Team A and Team B from pitch conditions

- **Training data:** pitch conditions and outcomes of previous matches
- **Testing data:** pitch conditions of upcoming matches
- **Data features:** historical pitch conditions, scores, match outcomes, and pitch conditions of upcoming matches

Sources of Data

- **Surveys:** ask a group of people questions (paper, interviews, online forms); e.g., population census every ten years
- **Web scraping:** collecting website data automatically with tools such as BeautifulSoup or Scrapy
- **Sensors:** measure temperature, pressure, light, sound, motion; e.g., smart homes, autonomous cars
- **Cameras:** images and video (CCTV, webcams); key to Computer Vision
- **Observations:** manual data gathering; e.g., counting birds in a season
- **APIs:** programs that fetch data from other services; e.g., social media posts for sentiment analysis
- **Kaggle:** datasets on text, images, tabular data; also competitions and notebooks (Python, R)



Acquiring Relevant Data from Reliable Sources

- Data must be authentic, reliable, and accurate; random websites may not be.
- Collection methods must be ethical: fair, lawful, and respectful of others' rights.
- Data should be open-sourced; accessing private data without permission is an offence.

Reliable source: government open-data portals

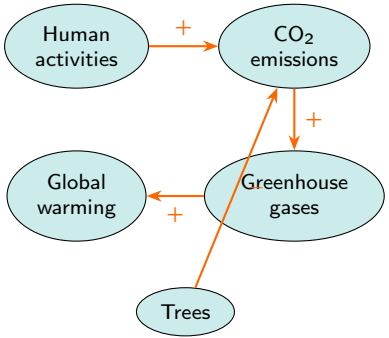
Examples: `data.gov.in` and `india.gov.in`.



System Maps

Visual diagrams showing the elements of an AI project and how they are connected.

- **Elements:** main parts working toward a purpose
- **Relationships:** how elements relate
- **Feedback loops:** cause-effect chains where output influences the system's own behaviour or inputs (positive or negative)
- Arrow $+$: directly related (X up $\Rightarrow$ Y up)
- Arrow $-$: inversely related (X up $\Rightarrow$ Y down)



Simplified sketch: trees absorb CO_2 (negative loop); human activities add it (positive loop).

Data Exploration

Need for Visualising Data and Tools

Visualisation helps to

- simplify complex data
- reveal trends, relationships, patterns
- uncover hidden relationships or anomalies
- select models for the next stage
- communicate insights, even to non-technical people

Tools

- **Tableau:** user-friendly, interactive, large datasets
- **Microsoft Power BI:** cloud BI, dashboards, reports
- **Qlik View:** shows data relationships by colours; in-memory speed
- **Looker Studio:** free web-based charts and dashboards
- **Excel:** manual; simple charts, not for large data
- **Plotly:** open-source Python; interactive 2D/3D graphs
- **Matplotlib:** Python; static, animated, interactive plots
- **Highcharts:** free JavaScript library; works on

Ways to Visualise Data

Chart	Description	Suitable for
Bar chart	Rectangular bars whose height/length is proportional to the values	Comparing categories; e.g., team scores, marks in subjects, rainy-day activity preferences
Line chart	Points connected by a line to track changes over time	Trends over time; e.g., bear population per year, temperature
Pie chart	Circle divided into sectors proportional to quantities	Contribution of each category to a whole; e.g., favourite fruit, energy sources, expense categories



Modelling

Deep Learning and Types of AI Models

- **Deep learning:** learns from large amounts of data using multi-layer neural networks that mimic the human brain; e.g., a driverless car identifying a person crossing the road.
- Humans understand graphs, but machines need data as numbers (0s and 1s).
- AI models define relationships between parameters *mathematically*.

Two approaches

Rule-based and **Learning-based**



Rule-based vs. Learning-based Approach

Rule-based

Developer feeds data plus rules ("If this happens, then do this").

- Banana/orange example: 200 labelled images; labels act as rules
- Unseen data (e.g., an apple) cannot be identified

Learning-based

Machine learns on its own from labelled and unlabelled data and defines its own rules.

- Example: 1000 mixed tree images clustered by height, colour, fruit, leaf type

Rule-based	**Learning-based**
Follows rules defined by the developer	Learns on its own from the data
AI via rule-based technique	AI via learning technique
Typically uses labelled data	Handles labelled and unlabelled data
May require less training time	Requires more training time

Supervised Learning and Regression

Supervised learning

- Trained on a labelled dataset (inputs and known outputs)
- Data split into training and test sets
- Fast training because results are already known
- Once trained, predicts the output for new input

Regression

- Predicts *continuous numerical values* (product price, salary)
- Predicts one variable from another
- Example: next day's temperature from humidity and cloud cover
- The learned function (a line on a graph) predicts future values



Unsupervised Learning

- Training input is neither classified nor labelled.
- The system does not know the correct output; it explores the data and finds the hidden (latent) structure.
- Once it learns the patterns, it can find them in new datasets.

Example

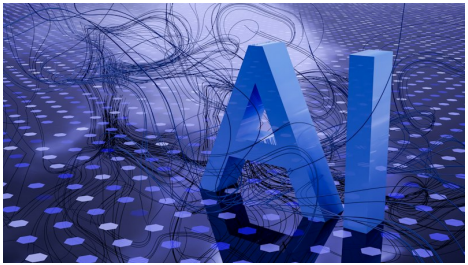
Fed images of a cycle, a car, and a helicopter, the model can separate them into groups, but it cannot say which cluster is the cycle because the data is unlabelled.



Evaluation and Deployment

AI Project Evaluation

- Assess the reliability and efficiency of a model by feeding it testing data and comparing output with the expected outcome.
- Testing data is separated from the acquired dataset at the data acquisition stage.
- Evaluation happens at various stages to find improvements and meet project goals.
- **Prediction:** forecasts about unknown or future data using patterns learned from training data.
- **Reality:** the actual situation when the prediction is made.



Prediction vs. Reality: Is It Raining?

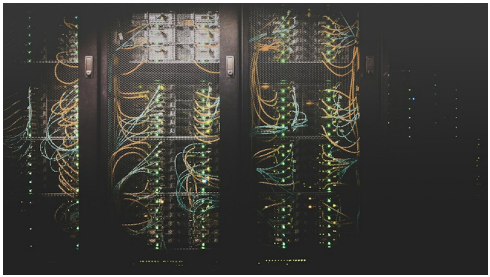
Rainfall model: is it raining at a given place?

	Reality: Yes	Reality: No
Pred.: Yes	True Positive	False Positive
Pred.: No	False Negative	True Negative

- **True Positive:** predicts rain; it is raining
- **True Negative:** predicts no rain; there is none
- **False Positive:** predicts rain; there is none
- **False Negative:** predicts no rain; it is raining

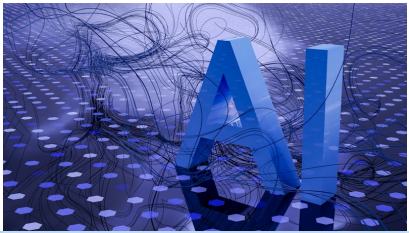
AI Project Deployment

- Implementing the AI model in a real-world setting.
- The model is integrated into the desired software or system and packaged for practical use.
- Goal: make the AI model useful for solving real problems.



Recap

- The AI project cycle has five stages: problem scoping, data acquisition, data exploration, modelling, and evaluation.
- The **4Ws canvas** and **Problem Statement Template** structure problem scoping.
- **Training data** trains the model; **testing data** checks its performance.
- **Data features** describe what information to collect; **system maps** show how elements relate.
- **Rule-based:** the machine follows developer-defined rules. **Learning-based:** the machine learns on its own.
- **Unsupervised learning** uses data that is neither classified nor labelled.



Thank You!

Questions?

